Assignment 3 : Cloud Auto-Scaling Using Local VM and GCP

1. Objective

The objective of this project is to design a system in which a local Virtual Machine continuously monitors its resource usage, specifically CPU utilization, and automatically offloads tasks to a cloud-based Virtual Machine when the CPU usage crosses a defined threshold of 75%. This helps in simulating a basic auto-scaling mechanism.

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2. Tools and Technologies Used

I used VirtualBox to make a machine on my computer and I chose Ubuntu as the operating system for this virtual machine (This part already created in assignment 1). I did the coding part using Python. I used Flask to create the API that's, on the cloud side. For CPU utilization % calcaulation, I have used psutil library.And for the local virtual machine and the cloud virtual machine to talk to each other I used the requests library.I used Google Cloud Platform to host the virtual machine.The cloud virtual machine is hosted on Google Cloud Platform.

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3. System Architecture

The system has two parts: a local virtual machine and a cloud virtual machine.The local virtual machine keeps an eye on CPU usage. Handles tasks on its own when everything is running smoothly.It checks the CPU usage all the time.When CPU usage gets too high it sends the task to the virtual machine.The cloud virtual machine has an API made with Flask that takes the task works on it and then sends the result back.The local virtual machine and the cloud virtual machine work together to get the job done.The cloud VM is where the task is processed when the local VM is busy.The task is sent to the VM to be processed.The result is then sent back, to the VM.

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4. Implementation Steps

In this assignment I used the machines I created in previous assignments. The local virtual machine I used was already set up in Assignment 1. The cloud virtual machine on Google Cloud Platform was created in Assignment 2. I did this because I wanted to save time and make sure everything worked the way.

For this project I made two Python programs: monitor.py and app.py. I put the monitor.py program on the virtual machine and it checks the CPU usage using the psutil library. It keeps checking the CPU usage. If it goes over 75 percent it sends a message to the cloud virtual machine. The cloud virtual machine runs the app.py program, which's a simple application that uses Flask. It gets the message does the work and then sends the answer back, to the system.

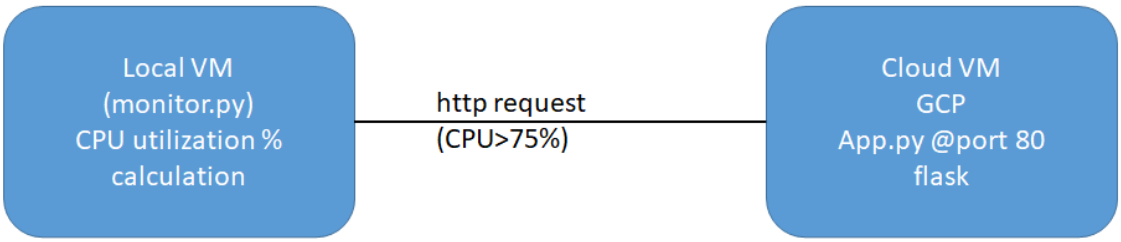

Figure 1 Architectrue